

ANALYSIS OF BI-METAL THERMOSTATS

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INTRODUCTION

The following investigation contains a general theory of bending of a bi-metal strip submitted to a uniform heating. This theory is applied in analysis of operation of a bi-metal strip thermostat. The equations are obtained for calculating the temperature of buckling, the complete travel during buckling, and the temperature of buckling in a backward direction. By using these equations the dimensions of the thermostat for a given temperature of operation and a given complete range of temperature can be calculated. The results obtained are based on certain ideal conditions. For example, it was assumed that the difference in the coefficients of expansion remained constant during heating, that the friction at the supports could be neglected and that the width

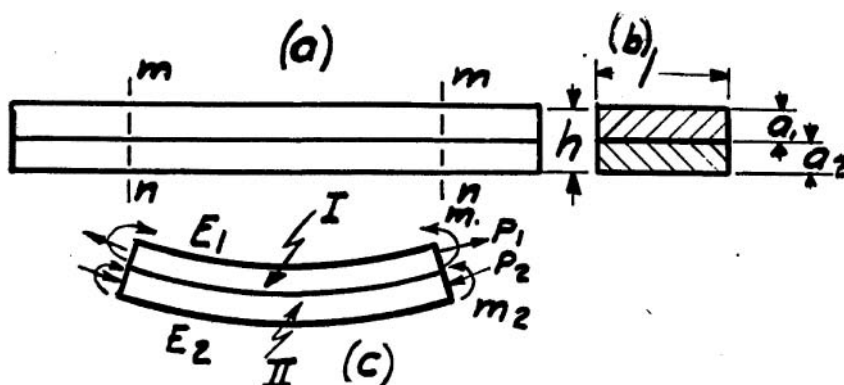


FIG. 1. Deflection of a bi-metal strip while uniformly heated.

of the strip could be considered as being very small. It is the opinion that these assumptions are not far removed from real conditions and the results obtained by using this theory can be considered as a useful guide in choosing the dimensions of thermostats and in discussing the stresses occurring in bi-metallic strips during operation.

1. *Deflection of a bi-metal strip while uniformly heated.* Let a narrow strip consisting of two metals welded together be uniformly heated from t_0° to $t^\circ\text{C}$. If the coefficient of linear expansion of these metals be different the heating will produce bending of the strip.

Let α_1 and α_2 denote the coefficients of expansion of the two metals (1) and (2), (see Fig. 1).

E_1 and E_2 denote their moduli of elasticity,

a_1 and a_2 their thicknesses,

h the thickness of the strip.

The width of the strip is taken as equal to unity. The following analysis has been made on the assumption that cross-sections of the strip originally plane and perpendicular to the axis remain plane during bending and become perpendicular to the curved axis of the strip. Consider an element cut out from the strip by two cross-sections mn and m_1n_1 . If $a_2 > a_1$, the deflection will be convex down, as shown in Fig. 1 (c). All the forces acting over the section of metal (1) on the concave side can be represented by an axial tensile force P_1 and bending moment M_1 . For metal (2) on the convex side all forces acting on the cross-section can be represented by an axial compressive force P_2 and bending moment M_2 . Due to the fact that there are not external forces acting on the strip, all forces acting over any cross-section of the strip must be in equilibrium, therefore,

$$P_1 = P_2 = P \quad (1)$$

$$\frac{Ph}{2} = M_1 + M_2 \quad (2)$$

Letting ρ = radius of curvature of strip,

$E_1 I_1$ = the flexural rigidity of the metal (1)

$E_2 I_2$ = the flexural rigidity of the metal (2), then,

$$M_1 = \frac{E_1 I_1}{\rho}$$

$$M_2 = \frac{E_2 I_2}{\rho}$$

Substituting in (2)

$$\frac{Ph}{2} = \frac{E_1 I_1 + E_2 I_2}{\rho} \quad (3)$$

Another equation for calculating P and ρ will be obtained from the consideration of deformation. On the bearing surface of both metals the unit elongation occurring in the longitudinal fibres of metals (1) and (2) must be equal, therefore,

$$a_1(t - t_0) + \frac{P_1}{E_1 a_1} + \frac{a_1}{2\rho} = a_2(t - t_0) - \frac{P_2}{E_2 a_2} - \frac{a_2}{2\rho} \quad \text{or}$$

using (1) and (3)

$$\frac{h}{2\rho} + \frac{2(E_1 I_1 + E_2 I_2)}{h\rho} \left(\frac{1}{E_1 a_1} + \frac{1}{E_2 a_2} \right) = (a_2 - a_1) (t - t_0)$$

From which,

$$\frac{1}{\rho} = \frac{(a_2 - a_1) (t - t_0)}{\frac{h}{2} + \frac{2(E_1 I_1 + E_2 I_2)}{h} \left(\frac{1}{E_1 a_1} + \frac{1}{E_2 a_2} \right)}$$

Letting,

$$\frac{a_1}{a_2} = m, \quad \frac{E_1}{E_2} = n,$$

and remembering that

$$I_1 = \frac{a_1^3}{12}, \quad I_2 = \frac{a_2^3}{12}$$

the following general equation for the curvature of a bi-metal strip will be obtained,

$$\frac{1}{\rho} = \frac{6(a_2 - a_1) (t - t_0) (1 + m)^2}{h \left(3(1 + m)^2 + (1 + mn) \left(m^2 + \frac{1}{mn} \right) \right)} \quad (4)$$

If the thicknesses of both metals are equal,

$$\begin{aligned} a_1 &= a_2, \quad m = 1 \\ \frac{1}{\rho} &= \frac{24(a_2 - a_1) (t - t_0)}{h(14 + n + 1/n)} \end{aligned} \quad (5)$$

The curvature is proportional to the difference in elongation of the two metals and inversely proportional to the thickness of the strip.

It is seen that the magnitude of the ratio $\frac{E_1}{E_2} = n$ does not produce any substantial effect on the curvature of the strip. Take for instance, $n = 1$, then,

$$\frac{1}{\rho} = \frac{3}{2} \frac{(a_2 - a_1) (t - t_0)}{h} \quad (6)$$

For $n = 1/2$ or $n = 2$,

$$\frac{1}{\rho} = \frac{48}{33} \frac{(a_2 - a_1)(t - t_0)}{h} \quad (7)$$

The difference between (6) and (7) is only about 3 per cent.

When the curvature $\frac{1}{\rho}$ is obtained the deflection δ of the strip can be easily evaluated. For instance, let a strip AB of bi-metal simply supported at the ends be bent due to uniform heating, as shown in Fig. 2. Then from the fact that the curve ACB is an arc of a circle having radius ρ it follows, that, $\delta(2\rho - \delta) = (l/2)^2$. Or remembering that the deflection δ is small in comparison with the radius of curvature we have,

$$2\rho\delta = \frac{l^2}{4},$$

from which,

$$\delta = \frac{l^2}{8\rho} \quad (8)$$

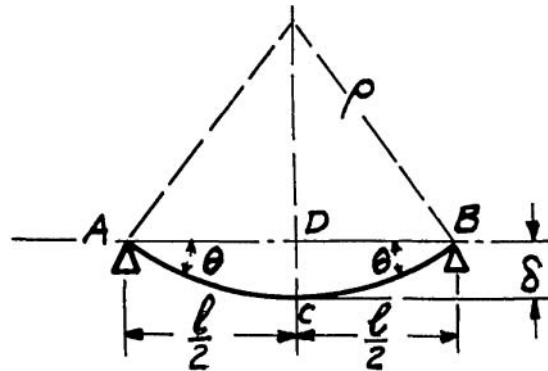


FIG. 2. Deflection of a simply supported bi-metal strip.

2. *Internal stresses during heating of a bi-metal strip.* The maximum stress during heating of bi-metal will be produced in the fibres on the bearing surface of the two metals. This stress consists of two parts: (a) the stress due to axial force and (b) the stress due to bending. For metal (1) it will be determined by the equation,

$$p_{max} = \frac{P}{a} + \frac{a_1 E_1}{2\rho}$$

or by using (3)

$$p_{max} = \frac{1}{\rho} \left(\frac{2}{ha_1} (E_1 I_1 + E_2 I_2) + \frac{a_1 E_1}{2} \right)$$

In the simplest case,

$$E_1 = E_2 = E$$

$$a_1 = a_2 = \frac{1}{2}h$$

$$p_{max} = \frac{E}{\rho} \frac{h}{4} \left(\frac{1}{3} + 1 \right), \text{ or using (b) ,}$$

$$p_{max} = \frac{E}{2} (a_2 - a_1) (t - t_0) \quad (9)$$

Letting $E = 27 \times 10^6$ lbs./sq. in.

$$a_2 - a_1 = 4 \times 10^{-6}$$

$$t - t_0 = 200^\circ\text{C}$$

the following is obtained,

$$p_{max} = 10800 \text{ lbs./sq. in.}$$

The stress distribution over the cross-section of the strip is shown in Fig. 3. It is easy to see that all stresses distributed over the cross-section are in equilibrium. They can be reduced to two equal couples of opposite directions.

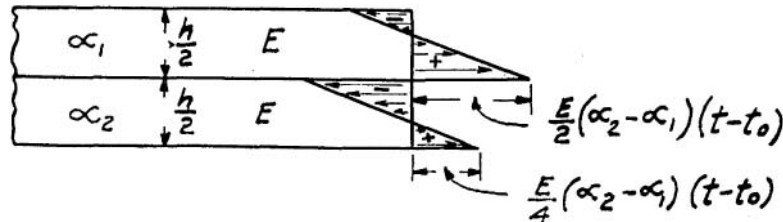


FIG. 3. Distribution of normal stresses on any cross section of bi-metal strip while uniformly heated.

In all previous discussion cross-sections remote from the ends of the strip were considered. Near the ends the picture of stress-distribution is more complicated. In the simplest case of two metals having equal thicknesses and equal moduli, shearing stresses between the two metals near the end will be produced during heating, as shown in Fig. 4. For every material these stresses can be reduced to a central force P , used in previous calculations, and to a couple $\frac{Ph}{4}$. The stresses produced by these forces are shown in Fig. 3 above. The distribution of shearing stress along the bearing surface cannot be determined in an elementary

way and it can be stated only that they are of "local" type and are concentrated near the ends of the strip along a distance, the magnitude of which is of the same order as the thickness h of the strip. This shows that the maximum intensity of these stresses may be of the same order as that of the normal stresses calculated above. This conclusion may be of some importance in considering the elastic property of material used for welding the two metals.

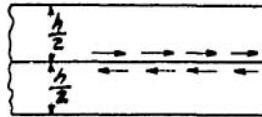


FIG. 4. Shearing stresses acting near the ends of bi-metal strip.

If the moduli and thicknesses of the two metals are different, not only shearing stresses but also normal stresses along the bearing surface near the ends of the strip will be produced during heating. These latter stresses will be also of local character and can be reduced to two equal and opposite couples of such magnitude as is necessary to produce, together with couples $\frac{Pa_1}{4}$ and $\frac{Pa_2}{4}$, previously considered, equal curvatures $1/\rho$ in both metals.

3. *Bending produced by external forces.* In this consideration the previous assumption that cross-sections, originally plane, remain so during the bending will be used. The problem on bending of a bar

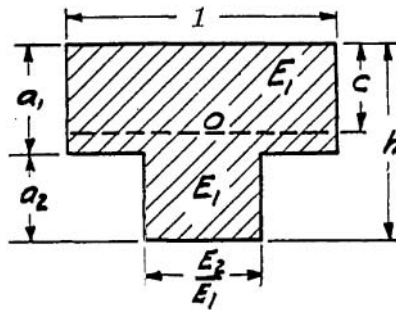


FIG. 5. Equivalent section of homogeneous strip.

consisting of two materials welded together can then be solved as follows.

Assume that $E_2 < E_1$ (Fig. 1). This fact can be taken into account by considering the bending of a strip of homogeneous material but of a section as shown in Fig. 5. The diminishing of the width corresponding

to the less rigid material must be taken in the ratio $\frac{E_1}{E_2}$ if the flexibility of strips has to remain the same. The stress distribution due to bending will depend on the position of the neutral axis going through the center of gravity O of the section. Let c determine the position of the center of gravity, then by using previous notation,

$$\frac{E_1}{E_2} = n, \quad \frac{a_1}{a_2} = m$$

$$c = \frac{\frac{(a_1+a_2)^2}{2} + \left(1 - \frac{1}{n}\right) \frac{a_1^2}{2}}{a_1 + \frac{a_2}{n}}$$

In the case, $a_1 = a_2 = h/2$

$$c = \frac{h(1 + 1/4(n-1))}{n+1} \quad (10)$$

The moment of inertia of the cross-section about the neutral axis will be,

$$I = \frac{(a_1+a_2)^3}{3n} + (1-1/n) \frac{a_1^3}{3} - \left(a_1 + \frac{a_2}{n}\right) c^2$$

In the case, $a_1 = a_2 = 1/2 h$,

$$I = h^3 \left(\frac{7}{24n} + \frac{1}{24} - \frac{\left(\frac{3}{n} + 1\right)^2}{32(1+1/n)} \right) \quad (11)$$

If both materials have the same modulus $n=1$ and from (11), we have

$$I = h^3 \left(\frac{1}{3} - \frac{1}{4} \right) = \frac{h^3}{12} = .0833 h^3$$

For $n = 1/.85$

$$I = h^3 (.2896 - .2128) = .0768 h^3 \quad (12)$$

For $n = 2$

$$I = .0573 h^3 \quad (13)$$

It is interesting to compare these results with the results obtained for the flexural rigidity of the strip, considering it as homogeneous and taking for the modulus of elasticity the average value $E = 1/2 (E_1 + E_2)$. Then,

$$EI = \frac{h^3 E_1}{12} \times \frac{1}{2} \left(1 + \frac{E_2}{E_1} \right) = \frac{h^3 E_1}{12} \times \frac{1+n}{2n} \quad (14)$$

For $n=1/.85$, equation (14) gives

$$EI = \frac{E_1 h^3}{1} \times \frac{1.85}{2} = .0772 E_1 h^3$$

This is in very good agreement with the equation (12).

For $n=2$,

$$EI = \frac{h^3 E_1}{12} \times \frac{3}{4} = .0625 E_1 h^3$$

As compared with (13) it gives a difference of about 9 per cent.

From this it can be concluded that in the calculation of flexural rigidity of a strip of bi-metal the formula for a homogeneous bar can be used with an average value of the modulus of elasticity. If n is near to unity (in the case of monel metal and nickel steel $n=1/.85$) this method of calculation is accurate enough.

By using this conclusion, the stresses produced in a constrained bi-metallic strip during heating can be easily obtained. Consider first

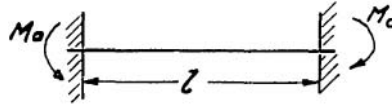


FIG. 6. Bi-metal strip with clamped ends.

a bi-metallic strip with built-in ends (Fig. 6). During uniform heating couples M_0 will occur at the ends. The magnitude of these couples must be such as to annihilate the curvature $1/\rho$ produced by heating (see Eq. 6). The equation for calculating M_0 becomes,

$$\frac{M_0}{EI} = 1/\rho = \frac{3}{2} \frac{(a_2 - a_1)(t - t_0)}{h}$$

In which,

$$E = \frac{E_1 + E_2}{2}$$

$$I = \frac{h^3}{12}$$

The corresponding maximum bending stress,

$$p = \frac{6 M_0}{h^2} = \frac{6}{h^2} \times \frac{3}{2} EI \frac{(a_2 - a_1)(t - t_0)}{h} = \frac{3}{4} E(a_2 - a_1)(t - t_0)$$

Combining the bending stresses due to M_0 with those produced in a free bi-metallic strip Fig. 3, the stress distribution shown in Fig. 7 will be obtained.

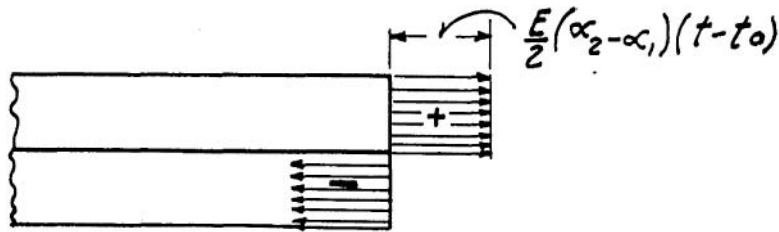


FIG. 7. Stress distribution in bi-metal strip with clamped ends.

In the case of a bi-metallic strip built in at the end A and having a pin at the end B (Fig. 8) the reactive force R must be such as to annihilate the deflection δ which would be produced by heating of a strip

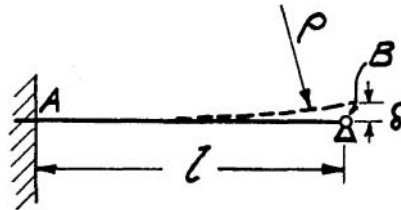


FIG. 8. Bending of bi-metal strip with one end clamped and another simply supported.

free at the end B . By using eq. 6 and the deflection formula for a cantilever beam, we have,

$$\frac{l^2}{2\rho} = \frac{Rl^3}{3EI}$$

from which,

$$R = \frac{3}{2} \frac{EI}{l} \times 1/\rho = \frac{9}{4} \frac{EI}{hl} (\alpha_2 - \alpha_1) (t - t_0)$$

The maximum bending moment at the built in end becomes

$$M_{max} = Rl = \frac{9}{4} \frac{EI}{h} (\alpha_2 - \alpha_1) (t - t_0)$$

and the corresponding maximum bending stress,

$$p_{max} = \frac{6 M_{max}}{h^2} = \frac{9}{8} E(\alpha_2 - \alpha_1) (t - t_0)$$

This stress is 50 per cent higher than that for the strip having both ends built in, which was obtained above.

4. *Deflection of a Bi-Metal Plate while uniformly heated.* If instead of a narrow strip a plate of bi-metal be heated, the curvatures in two different directions must be taken into consideration. For small deflections and with uniform heating these curvatures will be equal and constant over the surface of the plate, so that the plate becomes spherical in form. The radius of the curvature ρ can be obtained exactly in the same manner as shown above for the case of a narrow strip. It is necessary only to take into consideration that layers parallel to the neutral surface will suffer, due to spherical bending equal tension or compression in two perpendicular directions. Due to this fact, the equations of paragraph (1) can be used also in this case. It is necessary only to substitute $\frac{E_1}{1-\sigma}$ and $\frac{E_2}{1-\sigma}$ instead of E_1 and E_2 where σ denotes Poisson's ratio considered as equal for both materials. Due to the fact that the final equations (4) and (6) contain the ratio $\frac{E_2}{E_1}$ only, they will remain without any change and can be used in calculating the curvature during bending of the plates.

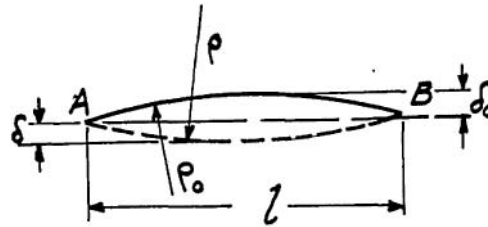


FIG. 9. Bending of curved bi-metal strip.

It is clear that the bending of the plate into a spherical form is impossible without some strain in the middle surface of the plate. If the deflections are small, say less than a half of the thickness of the plate, this strain in the middle surface of the plate can be neglected without any essential error, but with the increasing of the deflection, the effect of this strain becomes more and more pronounced and must be taken into consideration.

B. ANALYSIS OF THE STRIP TYPE OF BI-METAL THERMOSTAT

5. *Free bending of bi-metal strip.* Let a strip of bi-metal have an initial curvature equal to $1/\rho_0$ an initial deflection δ_0 (Fig. 9) and be bent in such a manner that the material with the larger coefficient of

expansion is on the concave side. During heating of this strip the curvature will gradually diminish and at a certain temperature the strip will pass through the plane form and afterwards become curved in a downward direction as shown by the dotted line in Fig. 9. The variation of the curvature will be obtained from equation (6)

$$\frac{1}{\rho_0} + \frac{1}{\rho} = \frac{3}{2} \frac{(a_2 - a_1)(t - t_0)}{h} \quad (15)$$

The corresponding deflection from (8) and (15) will be

$$\delta = \frac{l^2}{8\rho} = \frac{l^2}{8} \left(\frac{3}{2} \frac{(a_2 - a_1)(t - t_0)}{h} - \frac{1}{\rho_0} \right) = \frac{3}{16} \frac{l^2}{h} (a_2 - a_1)(t - t_0) - \delta_0 \quad (16)$$

For a thermostat a sudden buckling instead of gradual variation of curvature is essential and in order to obtain this characteristic fastening of the ends of the strip is necessary.

6. *Bi-metal curved strip in a rigid frame.* We will consider here the simplest case of a thermostat in the form of a curved bi-metal strip fastened between the fixed supports *A* and *B* in a rigid frame (see Fig. 10). It is assumed that the temperature expansions of the frame and

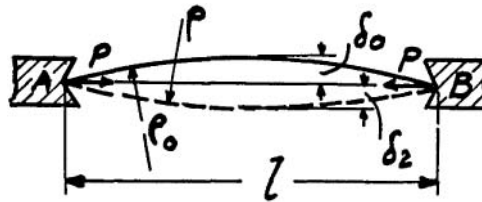


FIG. 10. *Bi-metal curved strip in a rigid frame.*

center line of the strip are equal and that there is no pressure of the strip on the supports at the initial temperature. Let l = length of the strip. The width is taken as equal to unity.

ρ_0 = the initial radius of curvature.

δ_0 = the initial deflection of the strip.

$A = 1 \times h$ = the area of the cross section of the strip.

$I = \frac{1 \times h^3}{12}$ = the moment of inertia of the cross section.

$E = \frac{1}{2}(E_1 + E_2)$ = the average modulus of elasticity.

δ = the deflection in a downward direction of the strip after heating for the case of "free ends" (see eq. 16).

δ_1 = the deflection measured in upward direction at which a sudden buckling of the strip begins.

δ_2 = the deflection of the strip in downward direction after buckling.

t_0 = the initial temperature of the strip.

t = the temperature at which sudden buckling occurs.

t_1 = the temperature when buckling in backward direction takes place.

$\Delta t = t - t_1$ = the range of the temperature.

t_2 = the temperature when the deflection for the case of "free ends" is equal to δ_0 .

The deformation of the strip during heating may be divided into two stages: first, a gradual decrease of the deflection from the initial value δ_0 to some value δ_1 , and secondly, a sudden buckling in a downward direction as shown by the dotted line in Fig. 10. During the gradual decrease of the initial curvature of the strip a longitudinal compression of the strip will occur. The corresponding compressive forces P tend to oppose buckling and may prevent it, provided the increase in temperature be not sufficiently great.

The temperature of buckling and the deflection δ_1 from which sudden buckling begins may be found if the following two expressions have been determined:

1. An expression giving the force P as a function of the decrease of the deflection of the strip during heating.

2. The magnitude of the compressive forces P necessary to prevent buckling of the strip when the temperature and the deflection are both given.

In order to obtain the increase in the compressive force due to decrease in the deflection, assume that the center line of the bent strip is a flat curve. Then the difference λ between the length of the curve and the length of the chord will be

$$\lambda = \frac{1}{2} \int_0^l \left(\frac{dy}{dx} \right)^2 dx \quad (17)$$

Without any great error it can be assumed that the initial deflection curve of the strip is a sine curve. Then we have,

$$y = \delta_0 \sin \frac{\pi x}{l}$$

Substituting in (17)

$$\lambda = \frac{\delta_0^2 \pi^2}{4l} \quad (18)$$

When, during heating, the deflection of the strip diminishes from δ_0 to δ_1 , the corresponding compression of the strip, from eq. (18), becomes,

$$\frac{\delta_0^2 \pi^2}{4l} - \frac{\delta_1^2 \pi^2}{4l}$$

and the compressive force will be,

$$P = \left(\frac{\delta_0^2 \pi^2}{4l} - \frac{\delta_1^2 \pi^2}{4l} \right) \frac{EA}{l} = \frac{EA \pi^2}{4l^2} \delta_0^2 \left(1 - \frac{\delta_1^2}{\delta_0^2} \right) \text{ or}$$

$$\frac{P l^2}{E I \pi^2} = \frac{3 \delta_0^2}{h^2} \left(1 - \frac{\delta_1^2}{\delta_0^2} \right) \quad (19)$$

This equation gives the force P as a function of the ratio $\frac{\delta_1}{\delta_0}$.

In evaluating the force necessary to prevent buckling of the strip, assume first that the supports A and B (see Fig. 10) are removed and the strip is free to assume during heating the form indicated by the line ACB in Fig. 11. The corresponding deflection δ may be calculated from equation (16). Assume now that this strip is bent in a direction convex up as shown by the dotted line and is held in this bent form by com-

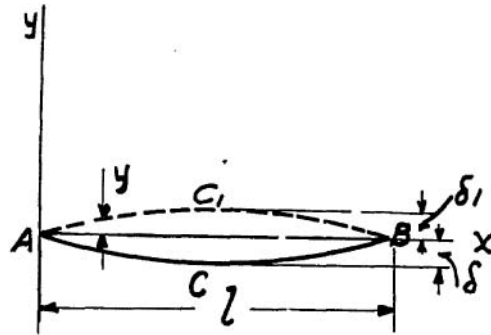


FIG. 11. Deflection curve of bi-metal strip.

pressive forces P . It is clear that the magnitude of the forces P will depend on the magnitude of δ_1 . In order to obtain the relation between δ_1 and P , assume that the initial curve ACB (Fig. 11) is a sine curve

$$\delta \sin \frac{\pi x}{l}$$

Then the displacements in the y direction occurring during bending from the shape ACB to the shape AC_1B are

$$y + \delta \sin \frac{\pi x}{l}$$

and the differential equation of the deflection curve AC_1B will be

$$EI \frac{d^2}{dx^2} \left(y + \delta \sin \frac{\pi x}{l} \right) = -Py$$

from which,

$$\frac{d^2 y}{dx^2} + \frac{P}{EI} y = \frac{\delta \pi^2}{l^2} \sin \frac{\pi x}{l} \quad (20)$$

Using notation,

$$\frac{P}{EI} = k^2 \quad (21)$$

we have,

$$y = A \sin kx + B \cos kx + \frac{\delta}{\frac{k^2 l^2}{\pi^2} - 1} \sin \frac{\pi x}{l} \quad (22)$$

where A and B are arbitrary constants of integration. Excluding the case when $\frac{kl}{\pi} = \text{an integer}^*$ it can be concluded from the conditions at the ends that $A = B = 0$.

Therefore,

$$\delta_1 = \frac{\delta}{\frac{k^2 l^2}{\pi^2} - 1}$$

or

$$\frac{Pl^2}{EI\pi^2} = \frac{\delta}{\delta_1} + 1 \quad (23)$$

Substituting for δ expression (16)

$$\frac{Pl^2}{EI\pi^2} = \left(\frac{3}{16} \frac{l^2}{h\delta_0} (a_2 - a_1) (t - t_0) - 1 \right) \frac{\delta_0}{\delta_1} + 1 \quad (24)$$

This equation gives the force P necessary to hold the strip ACB in the convex upwards shape AC_1B , shown by the dotted line in Fig. 11. It is seen that the magnitude of P increases with the decrease of the deflection δ_1 .

Equations (24) and (19) give the complete solution of the problem on buckling of the strip. If the force evaluated from (24) is always larger than that given by equation (19) this means that the reactions P at the supports A and B (Fig. 10) are insufficient to prevent buckling.

On the other hand, if for a certain value of the ratio $\frac{\delta_0}{\delta_1}$ the magnitude

* The corresponding values of P are given by the know Euler's column formula.

of P , as calculated from (24), becomes less than that obtained from equation (19), the reactions P of the frame will prevent buckling of the strip.

7. *Graphical Solution.* In order to determine the limit of temperature increase $t-t_0$, which is just necessary to produce the buckling of the strip, the following graphical method may be used.

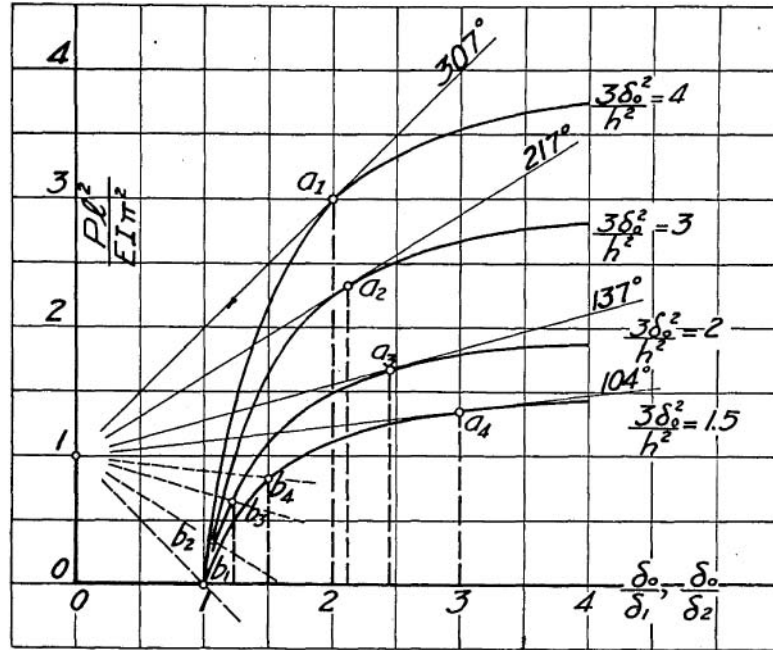


FIG. 12. Graphical determination of the temperature of operation of a bi-metal strip thermostat.

By applying equation (19), the curves representing $\frac{Pl^2}{EI\pi^2}$ as a function of $\frac{\delta_0}{\delta_1}$ are first plotted. Several curves of this kind, corresponding to various values of the ratio $\frac{3\delta_0^2}{h^2}$ are shown in Fig. 12. These curves have as their asymptotes horizontal lines distant $\frac{3\delta_0^2}{h^2}$ from the abscissa axis.

Furthermore, equation (24) may be represented graphically by straight lines going through the point 1 (Fig. 12) having the coordinates $x=0$, $y=\frac{Pl^2}{EI\pi^2}=1$. The slopes of these lines are given by the equation,

$$i = \frac{3}{16} \frac{l^2}{h\delta_0} (a_2 - a_1)(t - t_0) - 1 \quad (25)$$

It is seen that the slope depends on the temperature increase $t-t_0$. If the temperature increase is small, i also will be small and the straight line will cut the corresponding curve, given by equation (19). It means that at such a temperature no buckling occurs. The limiting temperature increase at which buckling becomes possible will be determined by the value of the slope of the tangent drawn through A to the corresponding curve, given by equation (19).

8. *Numerical Example.* Take, for instance, $\frac{h}{l} = 100$; $a_2 - a_1 = 4 \times 10^{-6}$ and the ratios $\frac{\delta_0}{h}$ of the initial deflection to the thickness of the strip are such as given by the second line of the table above. The corresponding curves given by equation (19) are plotted as previously dis-

TABLE A

$\frac{3\delta_0^2}{h^2} = 1.5$	2	3	4
$\frac{\delta_0}{h} = .707$	.814	1.000	1.154
$i = .111$	.272	.628	1.00
$t-t_0 = 104^\circ$	137°	217°	307°C
$\frac{\delta_1}{\delta_0} = .333$	.408	.472	.500
$\frac{\delta_2}{\delta_0} = .667$	.813	.940	1
$t_1-t_0 = 83^\circ$	79°	50°	0°
$\Delta t = 21^\circ$	58°	167°	307°

cussed. The values of the slopes of the tangents drawn through the point A to these curves are given in the third line of the table. Substituting these values in equation (25) the temperatures of buckling are obtained. These are given in the fourth line of the table.

It is seen that with increase of the initial deflection δ_0 , the temperature of operation increases. The slightest change in δ_0 produces a large effect on the temperature of buckling.

The abscissae of the points of tangency a_1 , a_2 , a_3 and a_4 (see Fig. 12) represent the values of the ratio $\frac{\delta_0}{\delta_1}$ at which sudden buckling of the strips commences. The corresponding values of $\frac{\delta_1}{\delta_0}$ are given in the fifth line of the table.

9. *Analytical Solution.* The same solution can be found also in an analytical way. By using notations,

$$\frac{Pl^2}{EI\pi^2} = y$$

$$\frac{\delta_0}{\delta_1} = x$$

$$\frac{3\delta_0^2}{h^2} = a$$

$$\frac{3}{16} \frac{l^2}{h\delta_0} (\alpha_2 - \alpha_1) (t - t_0) - 1 = i$$

we have, from equations (10) and (24),

$$y = a(1 - 1/x^2) \quad (19)^1$$

$$y = ix + 1 \quad (24)^1$$

The slope of the tangent to the curve (19)¹ is

$$\frac{dy}{dx} = \frac{2a}{x^3} \quad (26)$$

Substituting in (24)¹ expressions (26) and (19)¹ for i and y , the following expression for the abscissae of the point of tangency will be obtained,

$$x_0 = \frac{\delta_0}{\delta_1} = \sqrt{\frac{3a}{a-1}} \quad (27)$$

Substituting in this equation $a = 1.5, 2, 3$, and 4 , the reciprocal of the numbers given in the fifth line of the table above will be obtained. The slope of the tangent, from (26) and (27), becomes,

$$i = 2a(1/3 - 1/3a)^{3/2} \quad (28)$$

Substituting this in equation (25) we have

$$t - t_0 = \frac{1 + \frac{6\delta_0^2}{h^2} \left(\frac{1}{3} - \frac{1}{9} \frac{h^2}{\delta_0^2} \right)^{3/2}}{\frac{3}{16} \frac{l^2}{h\delta_0} (\alpha_2 - \alpha_1)} \quad (29)$$

From this equation the temperature of operation can be evaluated.

10. *Deflection of the strip after sudden buckling.* From the analysis given above it is seen that during heating of the strip the deflection,

initially equal to δ_0 , gradually decreases. When it attains the magnitude δ_1 , given by equation (27), a sudden buckling occurs and the strip, initially convex up (Fig. 10), becomes convex down. The differential equation of the deflection curve after buckling will be the same as equation (22) but the deflection will be now in downwards direction and its magnitude δ_2 will be given by the equation,

$$\delta_2 = \frac{\delta}{1 - \frac{kl^2}{\pi^2}}$$

Instead of equations (23) and (24) we will have

$$\frac{kl^2}{\pi^2} = \frac{Pl^2}{EI\pi^2} = 1 - \frac{\delta}{\delta_2} \quad (23)^1$$

$$\frac{Pl^2}{EI\pi^2} = 1 - \left(\frac{3}{16} \frac{l^2}{h\delta_0} (a_2 - a_1) (t - t_0) - 1 \right) \frac{\delta_0}{\delta_2} \quad (24)^1$$

Instead of equation (19) we will have,

$$\frac{Pl^2}{EI\pi^2} = \frac{3\delta_0^2}{h^2} \left(1 - \frac{\delta_2^2}{\delta_0^2} \right) \quad (19)^1$$

A very simple graphical solution of equations (19)¹ and (24)¹ is shown in Fig. 12. In order to obtain the magnitude of $\frac{\delta_0}{\delta_2}$ it is only necessary to draw through point I the lines $Ib_1, Ib_2 \dots$ having slopes,

$$-i = - \left(\frac{3}{16} \frac{l^2}{h\delta_0} (a_2 - a_1) (t - t_0) - 1 \right)$$

Then the abscissae of points $b_1, b_2 \dots$ give the values of $\frac{\delta_0}{\delta_2}$ and the

ordinates, the corresponding values of $\frac{Pl^2}{EI\pi^2}$. This solution enables

us to calculate very easily the displacements of the strip during sudden buckling. Take, for instance, the numerical example considered

above and assume $\frac{3\delta_0^2}{h^2} = 2$. Then from Fig. 12 and Table A, it is seen that the sudden buckling begins when,

$$\delta_1 = .408 \delta_0$$

The deflection in a downwards direction after buckling is equal to $\delta_2 = .813\delta_0$. The complete displacement during sudden buckling is,

$$\delta_1 + \delta_2 = 1.22 \delta_0$$

11. *Temperature at which Reverse Buckling Occurs.* If after sudden buckling of the strip the cooling begins, a certain limit t_1 for the lowering of the temperature may be established at which the strip becomes unstable and a sudden buckling in a backwards direction occurs so that the temperature controlled by the thermostat will vary within limits from t to t_1 . The range of temperatures $\Delta t = t - t_1$ is of practical importance and can be easily evaluated as follows.

Let t_2 denote the temperature necessary to produce in the strip with "free ends" the deflection δ equal to δ_0 (see equation (16)). At such a temperature the buckled strip will not press against the supports A and B of the frame, i.e., the conditions will be the same as in the initial unbuckled state. But now the material with the higher coefficient of expansion is on the convex side. Due to this fact the cooling will produce on the buckled strip the same effect as heating of the strip in its initial shape. The lowering of temperature $t_2 - t_1$ necessary to produce buckling in a backwards direction will be equal to the raising of the temperature $t - t_0$, calculated above (see table A). Therefore the temperature t_1 of the action of the thermostat in the backwards direction will be found from the equation,

$$t_2 - t_1 = t - t_0$$

from which

$$t_1 = t_2 - (t - t_0) \quad (30)$$

Here t_2 is determined from the equation,

$$\frac{3}{16} \frac{l^2}{h\delta_0} (\alpha_2 - \alpha_1) (t_2 - t_0) = 2 \quad (31)$$

By using equations (30) and (31) the numbers of the last two lines in the table A were obtained. It is seen from this table that with the lowering of the temperature of action the range Δt of temperatures diminishes, i.e., the thermostat becomes more sensitive to temperature variation.

From (25) and (31)

$$(t_2 - t_0) = \frac{2}{1+i} (t - t_0)$$

Substituting in (30)

$$l_1 - l_0 = \frac{1-i}{1+i}(l - l_0) \quad (32)$$

In order to obtain large sensitiveness the quantity i must be taken very small, i.e., $\frac{3\delta_0^2}{h^2}$ must be near to unity and δ_0 near to .577 h . Practically,

it is impossible to have sufficient accuracy with such a small initial deflection and in order to obtain sufficient sensitiveness with larger initial deflection the employment of a flexible frame becomes necessary.

12. *Bi-metal Curved Strip in a Flexible Frame.* Assume that due to the flexibility of the frame the reactive forces P (see Fig. 10) produce an increase Δ in the distance between the supports A and B . Then the compression of the strip produced by these forces during heating will be

$$\frac{\delta_0^2 \pi^2}{4l} - \frac{\delta_1^2 \pi^2}{4l} - \Delta$$

and the corresponding compressive force,

$$P = \left(\frac{\delta_0^2 \pi^2}{4l} - \frac{\delta_1^2 \pi^2}{4l} - \Delta \right) \frac{EA}{l} \quad (a)$$

In order to obtain the magnitude of Δ the flexibility of the frame must be taken into consideration. Assume that this flexibility is equivalent to the extensibility of a strip having the length l and the cross section A_1 . Then,

$$\Delta = \frac{Pl}{EA_1}$$

Substituting in (a)

$$P = k \cdot \frac{EA\pi^2}{4l^2} \delta_0^2 \left(1 - \frac{\delta_1^2}{\delta_0^2} \right) \quad \text{where,} \quad (33)$$

$$k = \frac{A_1}{A + A_1}$$

Instead of equation (19) we will have

$$\frac{Pl^2}{EI\pi^2} = k \cdot \frac{3\delta_0^2}{h^2} \left(1 - \frac{\delta_1^2}{\delta_0^2} \right) \quad (34)$$

Equation (24) remains without change. Now from (34) and (24) the temperature of operation can be obtained exactly in the same manner as explained above in the case of a rigid frame. The same graphical

method as shown in Fig. 12 can be used; it is necessary only to take the ordinates of the curves equal to

$$k \frac{3\delta_0^2}{h^2} \text{ instead of } \frac{3\delta_0^2}{h^2}$$

Take, for example, $k = 1/6$ and $\delta_0 = 2h$, then,

$$k \frac{3\delta_0^2}{h^2} = 2$$

The slope of the line Aa_3 in Fig. 12 will determine the temperature of operation. The magnitude of the slope, from Table A, is equal to .272 and equation (25) becomes

$$\frac{3}{16} \frac{l^2}{h\delta_0} (a_2 - a_1) (t - t_0) = 1.272$$

Substituting the previous data, we have,

$$t - t_0 = 338^\circ\text{C}$$

The temperature of backward buckling, from equation (32), becomes

$$t_1 - t_0 = \frac{1-i}{1+i} \left(t - t_0 \right) \frac{1-.272}{1+.272} \times 338 = 193^\circ\text{C}.$$

It is easy to see that a greater sensitiveness may be obtained by increasing the flexibility of the frame, i.e., by decreasing the slope i .

Applying equations (24), (32) and (34) the dimensions of the strip can be evaluated in such a manner so as to obtain, theoretically speaking, any given sensitiveness and any given temperature of operation. By using notations of the paragraph (8) these equations become,

$$\frac{t_1 - t_0}{t - t_0} = \frac{1-i}{1+i} \quad (32)^1$$

$$y = ka \left(1 - \frac{1}{x^2} \right) \quad (19)^{11}$$

$$y = ix + 1 \quad (24)^{11}$$

From (19),¹¹

$$\frac{dy}{dx} = \frac{2ka}{x^3} \quad (a)$$

Substituting (a) and (19)¹¹ in (24)¹¹

$$ka \left(1 - \frac{1}{x^2} \right) = \frac{2ka}{x^2} + 1$$

from which,

$$x_0 = \frac{\delta_0}{\delta_1} = \sqrt{\frac{3ka}{ka-1}} \quad (35)$$

Substituting in (a)

$$i = 2ka(\frac{1}{3} - 1/3ka)^{3/2} \quad (36)$$

Therefore,

$$t-t_0 = \frac{1+2ka(\frac{1}{3}-1/3ka)^{3/2}}{\frac{3}{16} \frac{l^2}{h\delta_0} (a_2-a_1)} \quad (37)$$

Assume, for instance, that the thermostat must have a temperature of operation $t-t_0=300^\circ$ and have such a sensitiveness that the backward buckling occurs when the lowering of the temperature becomes equal to 10 per cent of the temperature of operation, then, from (32)¹

$$\frac{1-i}{1+i} = .9 \text{ or } i = .053$$

Substituting in (36)

$$ka = k \frac{3\delta_0^2}{h^2} = 1.29 \quad (b)$$

Now, from (37),

$$\frac{l^2}{h\delta_0} = \frac{1+i}{\frac{3}{16}(t-t_0)(a_2-a_1)} = \frac{1.053}{\frac{3}{16} \times 300 \times 4 \times 10^{-2}} = \frac{4 \times 1.053 \times 10^4}{9}$$

Taking, for instance,

$$l/h = 200, \text{ we have,} \\ \delta_0 = 9/1.053h = 8.55h$$

From (35) we have,

$$\frac{\delta_0}{\delta_1} = 3.65 \quad (b)$$

From equations,

$$y \equiv ix = 1 \\ y = ka(1 - 1/x^2)$$

determining the deflection δ_2 of the strip after bending, we have,

$$\frac{\delta_0}{\delta_2} = 1.70 \quad (c)$$

The range of the complete travel during operation is

$$\delta_1 + \delta_2 = \delta_0 \left(\frac{1}{3.65} + \frac{1}{1.70} \right) = .86\delta_0 \quad (d)$$

It is seen that by increasing the ratio l/h the initial deflection and the range of displacement (d) can be increased.

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Refraktion und Akkommodation des menschlichen Auges,
Professor Dr. A. Siegrist, Director of the Eye Clinic in the University of Bern; pp. vi+148, with 108 illustrations, mostly colored. (Julius Springer, Berlin, 1925).

This exceedingly useful and attractive little volume is the outgrowth of a course of lectures on Refraction and Accommodation which Professor Siegrist has been in the habit of giving in the University of Bern. Apparently, therefore, the book is intended primarily for medical students in ophthalmology. The author is careful to state in the preface that he has taken particular pains to avoid all mathematical formulae; and so very little previous knowledge even of elementary geometrical optics is required in order to read the book. Nowadays when spectacle optics and methods of refraction are being developed in so many ways, one wonders if a skillful ophthalmologist can continue to be ignorant of such mathematics and physics as are almost necessary to gain a real insight into the intricate problems of the dioptrics of the eye and to ascertain the precise effect of using a lens or prism for the correction of the so-called anomalies of vision.

However, this book is evidently written by a practical ophthalmologist of wide experience and contains much valuable and authoritative information. Besides, it is beautifully illustrated, most of the pictures being colored. Some of them are of very great special interest. In the first ten pages there is an excellent modern description of the anatomy of the eye. This is followed by a chapter on the various types of spectacle glasses. Here the writer gives also an account of attempts that have been made to devise thin shell glasses to fit over the cornea and correct the curvature of this surface when it is very irregular. He describes also several forms of "hydrodiascope" for the same purpose, one of which he invented himself. The effect of accommodation on the intraocular pressure is discussed in the chapter on accommodation. The objections to Snellen's test charts for determining visual acuity are very clearly set forth, and various modern tests are described, especi-